17 · Case 13, part 2: contraception, and the three she hasn't answered

The words of this track, as it says them. Bold brown is the examiner.

Exam case 13, continued. Contraception, and the three starred questions she hasn't answered yet.

Three marks. Hormonal methods of contraception tend to affect a female's fertility by using progesterone. How do we use the normal functions of progesterone to prevent pregnancy?

The hook: stop ovulation, block the sperm, fake a pregnancy.

It prevents ovulation, by stopping LH from being released. It thickens the cervical mucus, which keeps the sperm out. And it imitates pregnancy, by keeping progesterone levels high.

Six marks. Contraception can be hormonal or non-hormonal. Summarise the differences in a table: the method, how it prevents pregnancy, and whether it prevents STIs.

The hook: hormones stop ovulation, a condom is a barrier, and only the condom stops STIs.

Hormonal methods. The pill, and the Jadelle implant, which slowly releases the hormones, may prevent ovulation from happening. The mini pill, which is progesterone only, prevents ovulation by stopping LH being released, thickens the cervical mucus to keep the sperm out, and imitates pregnancy by keeping progesterone high. Do hormonal methods prevent STIs? No. Pregnancy only. There's no barrier.

Non-hormonal methods. Condoms form a barrier that prevents the semen from getting into the woman's body. They protect against both pregnancy and STIs, like gonorrhoea, syphilis and HIV. But not pubic lice, which live in the pubic hair the condom doesn't cover. The IUD may prevent implantation of the conceptus in the endometrium, and it protects against pregnancy only.

Filing the IUD under non-hormonal is the Paper Sim's own reading, because some IUDs do carry hormones. But her session's point stands either way: only the condom also protects against STIs.

Now the three starred questions she hasn't posted answers to. They're still fair game in the exam, so here's a sound answer for each, built from her module content.

One. Male and female fertility are different. What are some of the key factors in male and female fertility?

For a male: enough sperm, that move well and are the right shape. The hormones that drive sperm production: FSH, LH and testosterone. And testes kept a little cooler than the rest of the body, which is why they sit in the scrotum. For a female: age, because the number and

quality of eggs fall over time. Regular ovulation, which needs FSH, LH, oestrogen and progesterone in balance. Open fallopian tubes, so the egg and sperm can meet. And a healthy uterine lining for implantation. For both: smoking, alcohol, weight, stress and sexually transmitted infections can all lower fertility. And for Rose: after Depo Provera, her cycle can take months to return.

Two. What are the functions of the sex hormones in the human body?

You already have this one, from Test 3. FSH grows the follicles in a female, and drives spermatogenesis in a male. LH triggers ovulation and forms the corpus luteum, and in a male it makes the Leydig cells produce testosterone. Oestrogen drives follicle growth, grows the reproductive organs and the breasts, lays fat on the hips, widens the pelvis, and rebuilds the endometrium in the proliferative phase. Progesterone readies the endometrium for implantation, in the secretory phase, and develops the breast alveoli in pregnancy. And testosterone drives spermatogenesis and the male secondary sex characteristics: more muscle, denser bone, and libido.

Three. Injections of the hormone FSH usually form part of IVF. What is IVF, when would it be appropriate, and what is the purpose of the FSH injections?

IVF is in vitro fertilisation. In vitro means in glass: the eggs are collected from the ovaries, fertilised by sperm in a lab, and the embryo is placed into the uterus. It's appropriate when the fallopian tubes are blocked or damaged, when the sperm count or movement is low, with ovulation problems, endometriosis, or unexplained infertility, or when simpler treatments haven't worked. The FSH injections stimulate the ovaries to grow several follicles at once, instead of the usual one a month. Her session says FSH is for the growth of the follicles. More follicles means more eggs to collect, and a better chance of a healthy embryo.

Once more. Contraception, and the three open questions.

Progesterone contraception stops ovulation by blocking LH, thickens the cervical mucus to keep sperm out, and imitates pregnancy. Hormonal methods, the pill, the Jadelle implant and the mini pill, prevent pregnancy only. Condoms are a barrier, and the only method that also stops STIs, though not pubic lice. The IUD may prevent implantation, and prevents pregnancy only. Fertility needs healthy sperm and hormones in a male, and regular ovulation, open tubes and a healthy lining in a female. IVF fertilises eggs in a lab and places the embryo in the uterus, and the FSH injections grow several follicles at once, so more eggs can be collected.